pipeline

April 14, 2025

1 Project Pipline

1.1 Setting Up Environment

1.1.1 Create Environment

```
module load miniconda3
conda activate
conda create --name project -y python=3.12
```

python-3.12.9

conda-25.3.0

```
conda activate project
```

1.1.2 Load Tools

```
module load OpenJDK/22.0.2
module load fastqc/0.12.1
module load samtools/1.21
module load nextflow/24.10.3
conda install -c bioconda -c conda-forge cutadapt -y
conda install -c bioconda star/2.7.10a -y
conda install -c bioconda multiqc -y
```

```
STAR --version
```

```
cd /courses/BINF6310.202530/students/SEC04_Group_3
```

load_pipeline_modules.sh This is for future convenience.

```
#!/bin/bash

echo "Loading available modules for the transcriptome pipeline..."

# Load modules that exist on Explorer
module load OpenJDK/22.0.2
module load perl/5.40.0
module load fastqc/0.12.1
module load samtools/1.21
module load nextflow/24.10.3
```

```
echo "Modules loaded:"
module list
```

check_software.sh This is to doublecheck for the future.

```
#!/bin/bash

# Exit if any command fails
set -e

# Install tools via conda/mamba if not already installed
echo "Installing required tools via conda..."
# conda install -y -c bioconda cutadapt

# Load system modules
module load fastqc/0.12.1
module load star/2.7.11b
module load samtools/1.21

# Confirm installation
echo ""
echo "  Installed tool versions:"
echo "-----------------------------"
fastqc --version
cutadapt --version
STAR --version
multiqc --version
featureCounts -v
echo "-----------------------------"
echo "All tools installed and ready!"
```

I have not installed featureCounts yet, so you should get that one error.

1.2 Trim

1.2.1 fastq_processing.nf

This will cutadapt fastq files in parallel

```
nextflow.enable.dsl = 2

params.quality = 20

// Define the input channel
fastqFiles = Channel.fromPath('./*.fastq')


// Define the trimming process
process trimAndFilter {
```

```
    input:
    path fastq

    output:
    path "trimmed/trimmed_${fastq.baseName}.fastq"

    script:
    """
    mkdir -p trimmed
    cutadapt -q ${params.quality} \\
             -o trimmed/trimmed_${fastq.baseName}.fastq \\
             ${fastq} > trimmed/${fastq.baseName}_cutadapt.log
    """
}

// Define the workflow
workflow {
    fastqFiles | trimAndFilter
}
```

```
nextflow run fastq_processing.nf
```

You should now see a **work** directory with a separate folder for each processed fastq file. If you drill down each of them, you will find a **trimmed** directory with the trimmed fastq version, ready for QC and then aligning.

In the parent directory:

```
mkdir -p trimmed_outputs
find work -name "trimmed_*.fastq" -exec cp {} trimmed_outputs/ \;

mkdir -p trimmed_outputs/logs
find work -name "*_cutadapt.log" -exec cp {} trimmed_outputs/logs/ \;
```

1.3 QC

In fastq folder:

```
mkdir qc_reports
fastqc *.fastq -o qc_reports/
```

You should see a separate html and zip file for each fastq processed. Now, time to consolidate them.

```
multiqc qc_reports/ -o qc_reports/qc_summary/
```

scp qc_summary to local system

scp -r explorer:/courses/BINF6310.202530/students/SEC04_Group_3/files/test/trimmed_outputs/qc_reports/q
.

1.4 STAR

Before continuing, make sure you are on a compute node and in your env.

```
srun --pty --partition=courses --export=ALL --mem=16G -t 4:00:00 bash
conda activate project
```

1.4.1 Download Reference Genome and Gene Transfer Format (GTF)

```
mkdir -p genome && cd genome
wget ftp://ftp.ensembl.org/pub/release-109/fasta/homo_sapiens/dna/Homo_sapiens.GRCh38.dna.prima

wget ftp://ftp.ensembl.org/pub/release-109/gtf/homo_sapiens/Homo_sapiens.GRCh38.109.gtf.gz

gunzip *
```

1.4.2 Build STAR Index

```
STAR --runThreadN 8 \
     --runMode genomeGenerate \
     --genomeDir ./genome_index \
     --genomeFastaFiles Homo_sapiens.GRCh38.dna.primary_assembly.fa \
     --sjdbGTFfile Homo_sapiens.GRCh38.109.gtf \
     --sjdbOverhang 100
```

or use build_star_index.sh This is what I ended up doing. It took a couple hours.

```
#!/bin/bash
#SBATCH --job-name=star_index             # Name of the job
#SBATCH --partition=courses               # Who to bill for the job
#SBATCH -N 1                              # How many nodes do you need. When in doubt, use 1
#SBATCH -c 16                             # How many "threads" do you need for your job
#SBATCH --mem 32G                         # How much memory
#SBATCH -t 8:00:00                        # How long
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=goodier.r@northeastern.edu
#SBATCH --out=/courses/BINF6310.202530/students/SEC04_Group_3/files/genome/genome_index/logs/%
#SBATCH --error=/courses/BINF6310.202530/students/SEC04_Group_3/files/genome/genome_index/logs/

# Setup environment
conda activate project

# Create logs directory if it doesn't exist
mkdir -p /courses/BINF6310.202530/students/SEC04_Group_3/files/genome/genome_index/logs

# Run STAR index generation
STAR --runThreadN 16 \
     --runMode genomeGenerate \
     --genomeDir ./genome_index \
     --genomeFastaFiles Homo_sapiens.GRCh38.dna.primary_assembly.fa \
```

```
    --sjdbGTFfile Homo_sapiens.GRCh38.109.gtf \
    --sjdbOverhang 100
```

```
sbatch build_star_index.sh
```

To monitor progress:

```
squeue -u $USER
cat genome_index/logs/star_index_113745.log
watch -n 300 'ls -lh genome_index1 | tail'
```

1.4.3 Nextflow Script

star_align.nf

```
nextflow.enable.dsl = 2

params.reads = '/scratch/goodier.r/project_files/trimmed_t2d/*.fastq'
// params.reads = '/scratch/goodier.r/project_files/trimmed_healthy/*.fastq'

params.outdir = '/scratch/goodier.r/project_files/trimmed_t2d/aligned_t2d'
// params.outdir = '/scratch/goodier.r/project_files/trimmed_healthy/aligned_healthy'

params.genomeDir = '/scratch/goodier.r/project_files/genome/genome_index'
params.threads = 8

Channel.fromPath(params.reads)
    .set { trimmed_fastqs }

process STAR_Align {
    input:
        tuple path(fastq), val(genomeDir)

    output:
        path "*.bam"

    script:
        sample_id = fastq.getBaseName().replaceFirst(/^trimmed_/, "").replaceFirst(/\.fastq$/,
        """
        STAR --genomeDir $genomeDir \
            --readFilesIn $fastq \
            --runThreadN ${params.threads} \
            --outTmpDir ${sample_id}_STARtmp \
            --outSAMtype BAM SortedByCoordinate \
            --outFileNamePrefix ${sample_id}_

        # Rename STAR output to something simpler
        mv ${sample_id}_Aligned.sortedByCoord.out.bam ${sample_id}.bam

        # Copy results to final output dir
```

```
        mkdir -p ${params.outdir}
        cp ${sample_id}.bam ${params.outdir}/

        cp -f ${sample_id}.Log.final.out ${params.outdir}/ || true
        """
}

workflow {
    trimmed_fastqs
        .combine(Channel.value(params.genomeDir))
        | STAR_Align
}
```

```
nextflow info
```

```
  Version: 24.10.5 build 5935
  Created: 04-03-2025 17:55 UTC (12:55 GMT-04:00)
  System: Linux 5.14.0-362.13.1.el9_3.x86_64
  Runtime: Groovy 4.0.23 on OpenJDK 64-Bit Server VM 21.0.6+7-LTS
  Encoding: UTF-8 (UTF-8)
```

```
nextflow run star_align.nf
```

This took about 3:30 hours for all 149 fastq files.

1.5 featureCounts

Before continuing, make sure you are on a compute node and in your env.

```
srun --pty --partition=courses --export=ALL --mem=16G -t 4:00:00 bash
conda activate project
```

1.5.1 Nextflow Script

featureCounts.nf

```
nextflow.enable.dsl = 2

params.bam_dir = '/scratch/nagajothipurushota.a/trimmed_outputs_healthy/trimmed_outputs/alignec
params.outdir = '/scratch/nagajothipurushota.a/trimmed_outputs_healthy/trimmed_outputs/featureC
params.gtf = '/scratch/nagajothipurushota.a/trimmed_outputs_healthy/trimmed_outputs/Homo_sapier
params.threads = 8
params.memory = '64 GB'

Channel
    .fromPath(params.bam_dir, checkIfExists: true)
    .ifEmpty { error "No BAM files found at: ${params.bam_dir}" }
    .collect()
    .set { aligned_bams }

process FeatureCounts {
```

```
    input:
        path bam_files

    output:
        path "counts.txt"
        path "counts.txt.summary"

    conda "~/.conda/envs/project"
    memory params.memory
    cpus params.threads

    script:
    """
    featureCounts \\
        -a ${params.gtf} \\
        -t exon \\
        -g gene_id \\
        -p \\
        -T ${params.threads} \\
        -o counts.txt \\
        ${bam_files.join(' ')}

    mkdir -p ${params.outdir}
    cp counts.txt ${params.outdir}/
    cp counts.txt.summary ${params.outdir}/
    """
}

workflow {
    aligned_bams | FeatureCounts
}
```

####nextflow.config

```
process {
    executor = 'slurm'
    queue = 'short'
    memory = '64 GB'
    cpus = 8
}
```

To run the featureCounts script

```
nextflow run /scratch/nagajothipurushota.a/trimmed_outputs_healthy/trimmed_outputs/featureCount
```

Output files -> * counts.txt
* counts.txt.summary

1.6 Transform Count Matrix into RPKM Values

Using an R session:

```
if (!requireNamespace("BiocManager", quietly = TRUE))
    install.packages("BiocManager")
BiocManager::install("edgeR")

library(edgeR)

counts <- read.table("/scratch/goodier.r/counts_combined.txt", header = TRUE, row.names = 1)

gene_lengths <- counts$Length

count_matrix <- counts[, !(colnames(counts) %in% c("Geneid", "Chr", "Start", "End", "Strand",

y <- DGEList(counts = count_matrix)
rpkm_values <- rpkm(y, gene.length = gene_lengths)

write.table(rpkm_values, "rpkm_combined.txt", sep = "\t", quote = FALSE)
```

analysis

April 11, 2025

1 Analysis of RPKM Tables

[1]:
```
import os
import numpy as np
import pandas as pd

import mygene

import matplotlib.pyplot as plt
import plotly.express as px


rpkm = r"../rpkm_values/rpkm_combined.txt"
healthy = r"../rpkm_values/rpkm_healthy/rpkm_values_healthy.txt"
t2d = r"../rpkm_values/rpkm_t2d/rpkm_values_t2d.txt"
```

[2]:
```
df = pd.read_csv(rpkm, sep='\t')
df_healthy = pd.read_csv(healthy, sep='\t')
df_t2d = pd.read_csv(t2d, sep='\t')
```

[3]:
```
# df.columns.tolist()
```

[4]:
```
print(len(df.columns))
print(len(df_healthy.columns))
print(len(df_t2d.columns))
```

```
142
97
45
```

[5]:
```
# Transpose the RPKM dataframe so rows become samples
df_t = df.T

# Now determine sample labels
healthy_samples = set(df_healthy.columns)
t2d_samples = set(df_t2d.columns)

# Assign condition based on sample name
```

```
df_t['condition'] = ['healthy' if sample in healthy_samples else 't2d' for␣
 ↪sample in df_t.index]
df_t.head()
```

[5]:

```
            ENSG00000160072  ENSG00000279928  ENSG00000228037  \
AZ_A8.bam          0.000000              0.0              0.0
AZ_C10.bam         0.000000              0.0              0.0
AZ_C4.bam          0.000000              0.0              0.0
AZ_D6.bam          0.000000              0.0              0.0
AZ_E7.bam          0.073673              0.0              0.0

            ENSG00000142611  ENSG00000284616  ENSG00000157911  \
AZ_A8.bam               0.0              0.0              0.0
AZ_C10.bam              0.0              0.0              0.0
AZ_C4.bam               0.0              0.0              0.0
AZ_D6.bam               0.0              0.0              0.0
AZ_E7.bam               0.0              0.0              0.0

            ENSG00000269896  ENSG00000228463  ENSG00000260972  \
AZ_A8.bam               0.0              0.0              0.0
AZ_C10.bam              0.0              0.0              0.0
AZ_C4.bam               0.0              0.0              0.0
AZ_D6.bam               0.0              0.0              0.0
AZ_E7.bam               0.0              0.0              0.0

            ENSG00000224340  …  ENSG00000276345  ENSG00000273532  \
AZ_A8.bam               0.0  …          0.000000              0.0
AZ_C10.bam              0.0  …          0.000000              0.0
AZ_C4.bam               0.0  …          0.000000              0.0
AZ_D6.bam               0.0  …          4.866315              0.0
AZ_E7.bam               0.0  …          0.000000              0.0

            ENSG00000275063  ENSG00000277856  ENSG00000271254  \
AZ_A8.bam               0.0              0.0         0.000000
AZ_C10.bam              0.0              0.0         0.000000
AZ_C4.bam               0.0              0.0         0.000000
AZ_D6.bam               0.0              0.0         0.000000
AZ_E7.bam               0.0              0.0         0.131931

            ENSG00000275987  ENSG00000268674  ENSG00000277475  \
AZ_A8.bam               0.0              0.0              0.0
AZ_C10.bam              0.0              0.0              0.0
AZ_C4.bam               0.0              0.0              0.0
AZ_D6.bam               0.0              0.0              0.0
AZ_E7.bam               0.0              0.0              0.0

            ENSG00000275405  condition
```

```
AZ_A8.bam                0.0    healthy
AZ_C10.bam               0.0    healthy
AZ_C4.bam                0.0    healthy
AZ_D6.bam                0.0    healthy
AZ_E7.bam                0.0    healthy

[5 rows x 62711 columns]
```

[6]:
```
df_t['donor'] = df_t.index.str.split('_').str[0]
```

1.1 Beta-cell Confirmation

[7]:
```
mg = mygene.MyGeneInfo()

# Query gene symbols
gene_symbols = ['GCG', 'INS', 'PPY', 'SST']
results = mg.querymany(gene_symbols, scopes='symbol', fields='ensembl.gene',␣
 ↪species='human')

# Build dict symbol -> Ensembl ID
symbol_to_ensembl_small = {r['query']: r['ensembl']['gene'] if␣
 ↪isinstance(r['ensembl'], dict) else r['ensembl'][0]['gene']
                     for r in results if 'ensembl' in r}
```

```
Input sequence provided is already in string format. No operation performed
Input sequence provided is already in string format. No operation performed
```

[8]:
```
symbol_to_ensembl_small
```

[8]:
```
{'GCG': 'ENSG00000115263',
 'INS': 'ENSG00000254647',
 'PPY': 'ENSG00000108849',
 'SST': 'ENSG00000157005'}
```

[9]:
```
# Create a subset with just those 4 genes
subset = df_t[list(symbol_to_ensembl_small.values())].copy()
subset.columns = symbol_to_ensembl_small.keys()  # Rename to gene symbols
subset
# Normalize by log2
log_subset = np.log2(subset + 1)
log_subset
```

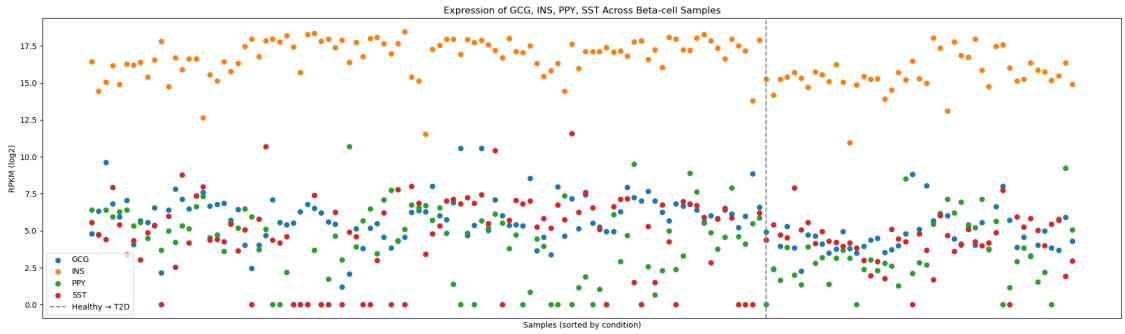
[9]:

	GCG	INS	PPY	SST
AZ_A8.bam	4.787959	16.437794	6.413168	5.555190
AZ_C10.bam	5.160662	15.960881	1.175719	6.234554
AZ_C4.bam	5.275064	17.133383	1.060861	5.122213
AZ_D6.bam	4.657026	14.429525	7.356343	5.765354
AZ_E7.bam	7.975794	16.332966	0.000000	6.733543

```
…                          …         …         …         …
HP1526901T2D_N12.bam  4.358434  15.255884  3.031123  1.978236
HP1526901T2D_N13.bam  4.501219  15.279048  2.316015  2.911368
HP1526901T2D_N23.bam  3.546375  13.924297  2.808932  1.791585
HP1526901T2D_N3.bam   8.056463  14.997494  2.688993  3.690260
HP1526901T2D_O14.bam  4.318461  14.894209  5.084971  2.954143

[142 rows x 4 columns]
```

[10]:
```
# Sort df_t by condition
df_sorted = df_t.sort_values(by='condition')

# Get just the log2(RPKM + 1) expression of selected genes, sorted by sample␣
↪order
log_subset_sorted = np.log2(df_sorted[list(symbol_to_ensembl_small.values())] +␣
↪1)
log_subset_sorted.columns = symbol_to_ensembl_small.keys()

transition_idx = df_sorted['condition'].eq('t2d').idxmax()
transition_pos = df_sorted.index.get_loc(transition_idx)
```

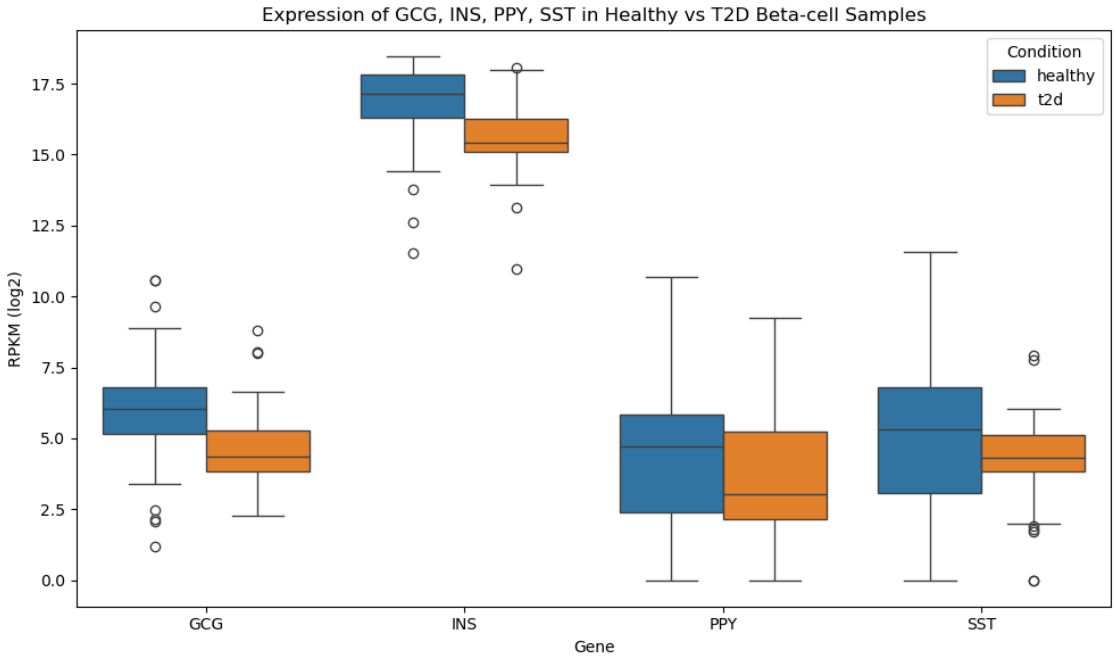
[11]:
```
plt.figure(figsize=(20, 6))

for gene in ['GCG', 'INS', 'PPY', 'SST']:
    plt.scatter(log_subset_sorted.index, log_subset_sorted[gene], label=gene)

# Add vertical dotted line at transition
plt.axvline(x=transition_pos, linestyle='--', color='gray', label='Healthy →␣
↪T2D')

# Simplify x-axis
plt.xticks([], [])  # Hide x-tick labels
plt.ylabel('RPKM (log2)')
plt.xlabel('Samples (sorted by condition)')
plt.title('Expression of GCG, INS, PPY, SST Across Beta-cell Samples')
plt.legend()
plt.tight_layout()
plt.show()
```



[21]:
```
# Melt the DataFrame
long_df = log_subset_sorted[['GCG', 'INS', 'PPY', 'SST']].copy()
condition_labels = pd.Series(condition_labels, index=log_subset_sorted.index)
long_df['condition'] = condition_labels
long_df = long_df.melt(id_vars='condition', var_name='Gene',␣
↪value_name='Expression')
```

[ ]:
```
condition_labels
```

[22]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
sns.boxplot(x='Gene', y='Expression', hue='condition', data=long_df)
plt.ylabel('RPKM (log2)')
plt.title('Expression of GCG, INS, PPY, SST in Healthy vs T2D Beta-cell␣
↪Samples')
plt.legend(title='Condition')
plt.tight_layout()
plt.show()
```

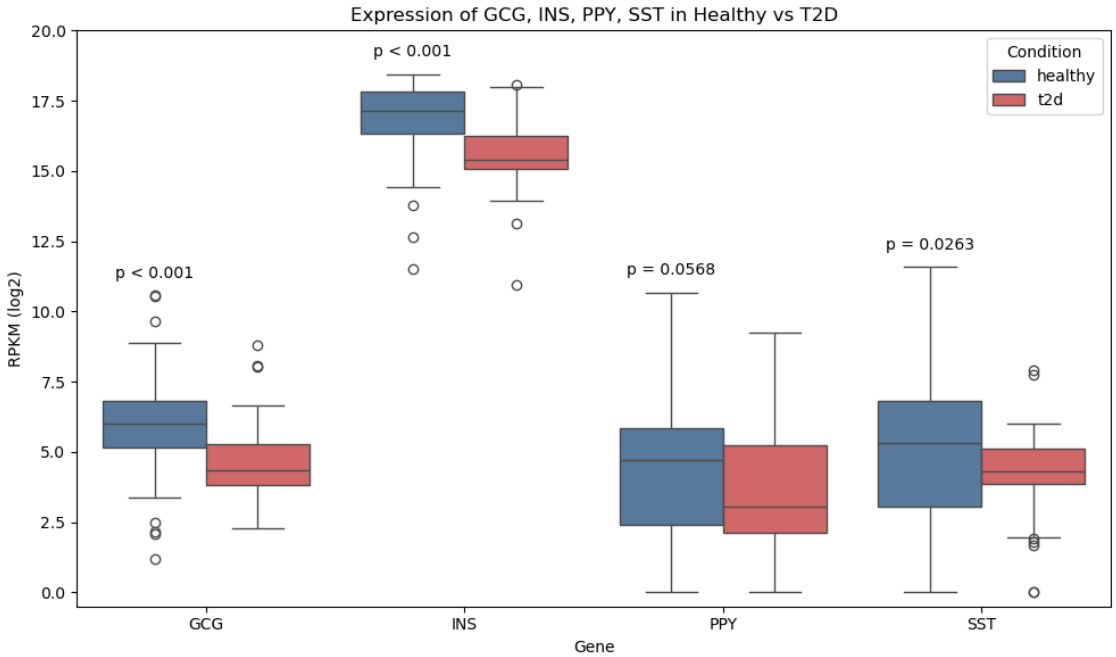


[30]:
```
from scipy.stats import mannwhitneyu

genes = ['GCG', 'INS', 'PPY', 'SST']
results = []

for gene in genes:
    healthy_values = log_subset_sorted.loc[condition_labels == 'healthy', gene].
 ↪dropna()
    t2d_values = log_subset_sorted.loc[condition_labels == 't2d', gene].dropna()

    if len(healthy_values) == 0 or len(t2d_values) == 0:
        print(f"{gene}: One group is empty. Skipping.")
        continue

    stat, p = mannwhitneyu(healthy_values, t2d_values, alternative='two-sided')
    results.append((gene, stat, p))

# Display results
for gene, stat, p in results:
    print(f"{gene}: U-statistic={stat:.2f}, p-value={p:.4f}")

print(results)
```

```
GCG: U-statistic=3367.00, p-value=0.0000
INS: U-statistic=3438.00, p-value=0.0000
PPY: U-statistic=2617.00, p-value=0.0568
```

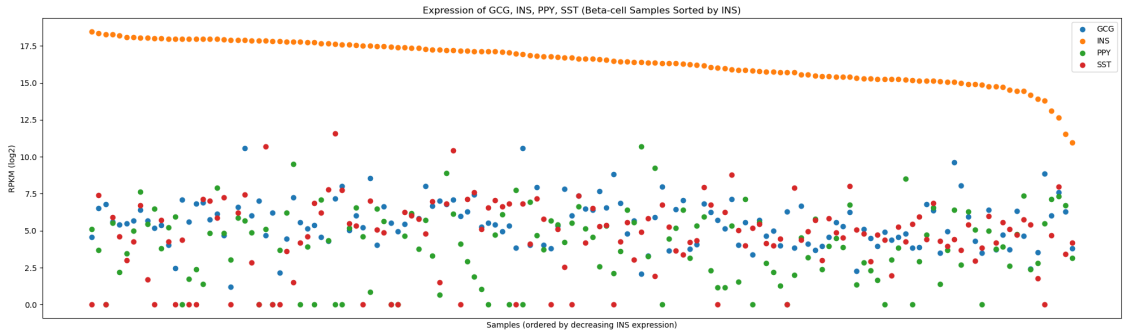
```
SST: U-statistic=2689.00, p-value=0.0263
[('GCG', 3367.0, 2.0874964229162652e-07), ('INS', 3438.0,
3.740623914069236e-08), ('PPY', 2617.0, 0.05675463792123318), ('SST', 2689.0,
0.0262698723679572)]
```

[38]:
```python
# P-values you already computed
pvals = {
    'GCG': 0.0000,
    'INS': 0.0000,
    'PPY': 0.0568,
    'SST': 0.0263,
}

custom_palette = {'healthy': '#4E79A7', 't2d': '#E15759'}

plt.figure(figsize=(10, 6))
ax = sns.boxplot(x='Gene', y='Expression', hue='condition', data=long_df,␣
↪palette=custom_palette)
plt.ylabel('RPKM (log2)')
plt.title('Expression of GCG, INS, PPY, SST in Healthy vs T2D')

# Add p-value annotations
x_offset = {'GCG': -0.2, 'INS': -0.2, 'PPY': -0.2, 'SST': -0.2}
for i, gene in enumerate(['GCG', 'INS', 'PPY', 'SST']):
    y_max = long_df[long_df['Gene'] == gene]['Expression'].max()
    pval = pvals[gene]
    text = f"p = {pval:.4f}" if pval >= 0.001 else "p < 0.001"
    plt.text(i + x_offset[gene], y_max + 0.5, text, ha='center', va='bottom',␣
↪fontsize=10)

plt.ylim((-0.5, 20))
plt.legend(title='Condition')
plt.tight_layout()
plt.show()
```



[12]:
```
# Sort by INS decreasing
ensg_ins = symbol_to_ensembl_small['INS']
df_sorted = df_t.sort_values(by=ensg_ins, ascending=False)
log_subset_sorted = np.log2(df_sorted[list(symbol_to_ensembl_small.values())] +␣
↪1)
log_subset_sorted.columns = symbol_to_ensembl_small.keys()  # rename to gene␣
↪symbols
```

[13]:
```
plt.figure(figsize=(20, 6))

for gene in ['GCG', 'INS', 'PPY', 'SST']:
    plt.scatter(log_subset_sorted.index, log_subset_sorted[gene], label=gene)

plt.xticks([], [])  # Hide sample names
plt.ylabel('RPKM (log2)')
plt.xlabel('Samples (ordered by decreasing INS expression)')
plt.title('Expression of GCG, INS, PPY, SST (Beta-cell Samples Sorted by INS)')
plt.legend()
plt.tight_layout()
plt.show()
```



1.2 t-SNE (t-distributed Stochastic Neighbor Embedding)

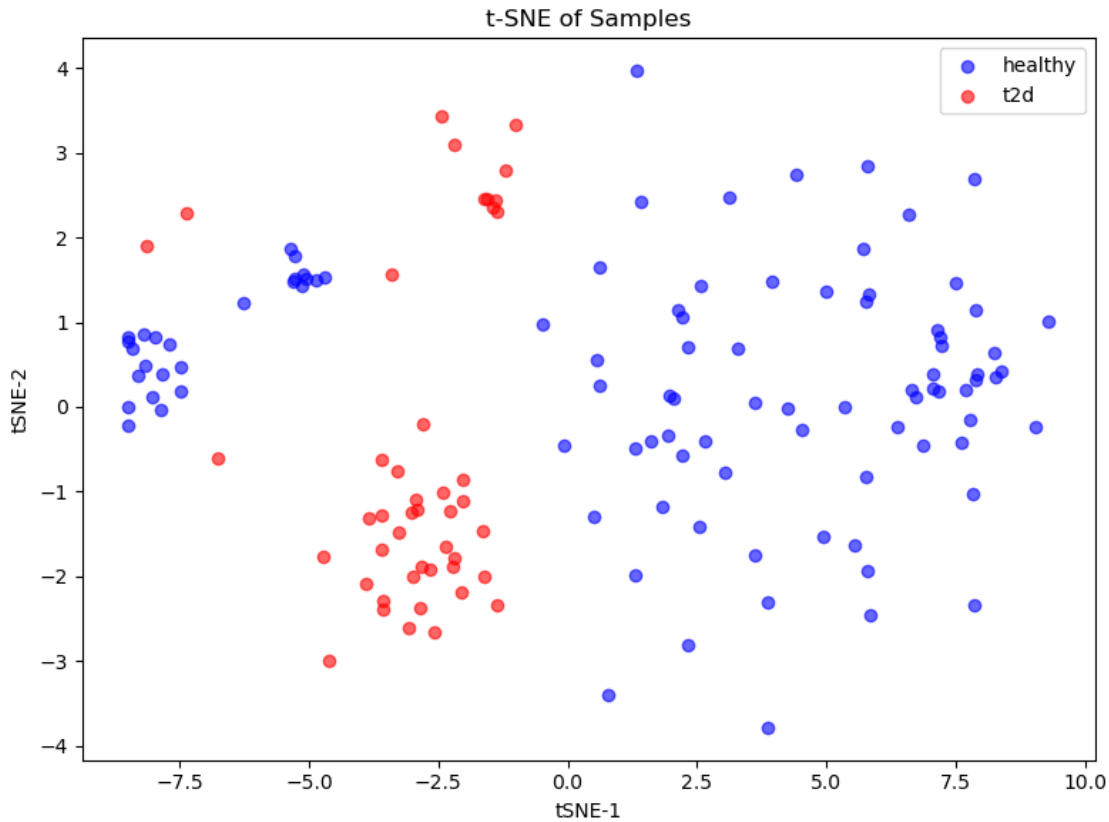
1.2.1 All 62710 Genes

[14]:
```
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
```

[73]:
```
# Drop the label column to get expression matrix
X = df_t.drop(columns=['condition', 'donor'])
X_log = np.log2(X + 1)
```

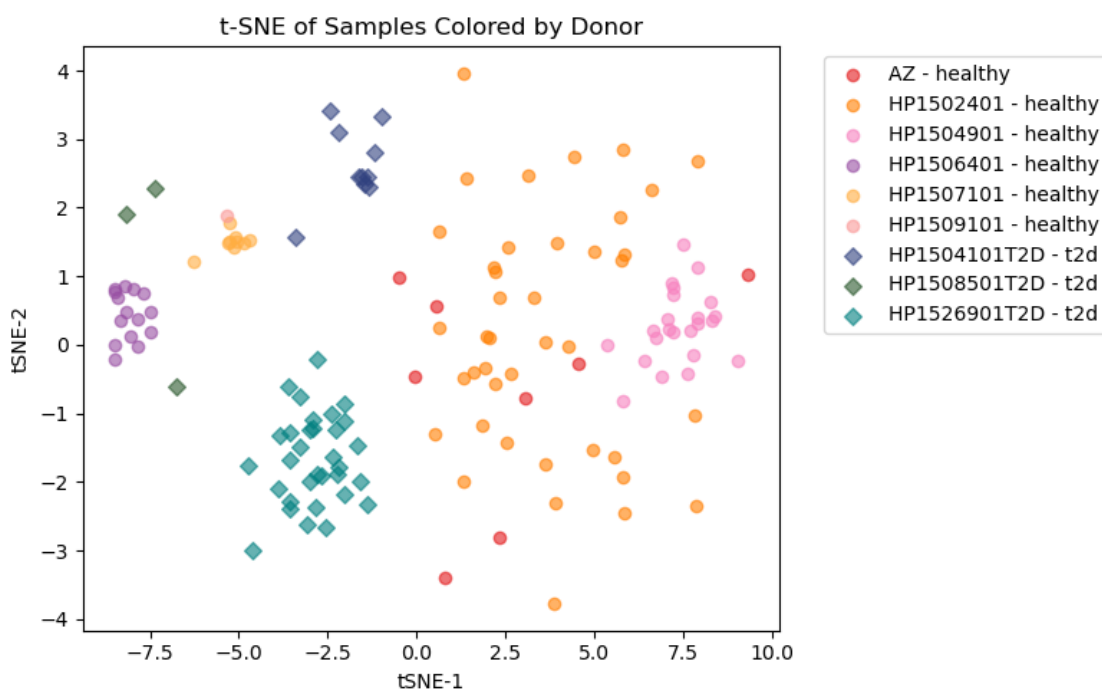
[16]:
```
# PCA
X_pca = PCA(n_components=30).fit_transform(X_log)
# Run t-SNE
X_tsne = TSNE(n_components=2, random_state=42).fit_transform(X_pca)
```

[17]:
```
# Plot
plt.figure(figsize=(8, 6))
colors = {'healthy': 'blue', 't2d': 'red'}
for cond in df_t['condition'].unique():
    ix = df_t['condition'] == cond
    plt.scatter(X_tsne[ix, 0], X_tsne[ix, 1], label=cond, c=colors[cond],␣
 ↪alpha=0.6)

plt.title('t-SNE of Samples')
plt.xlabel('tSNE-1')
plt.ylabel('tSNE-2')
plt.legend()
plt.tight_layout()
plt.show()
```



[18]:
```
# Plot with donor-specific colors
plt.figure(figsize=(8, 5))
markers = {'healthy': 'o', 't2d': 'D'}
colors = [
    "#e41a1c",  # warm - red
    "#ff7f00",  # warm - orange
    "#f781bf",  # warm - pink
    "#984ea3",  # warm - purple
    "#ffae42",  # warm - goldenrod
    "#fb9a99",  # warm - salmon

    "#30417f",  # cool - blue
    "#2b592e",  # cool - green
    "#008080"   # cool - teal
]
donors = df_t['donor'].unique()
donor_colors = dict(zip(donors, colors))

for donor in df_t['donor'].unique():
    ix = df_t['donor'].values == donor
```

```
    condition = df_t[df_t['donor'] == donor]['condition'].unique()[0]

    plt.scatter(
        X_tsne[ix, 0],
        X_tsne[ix, 1],
        label=f"{donor} - {condition}",
        color=donor_colors[donor],
        marker=markers.get(condition, 'o'),
        alpha=0.6
    )

plt.title('t-SNE of Samples Colored by Donor')
plt.xlabel('tSNE-1')
plt.ylabel('tSNE-2')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



1.2.2 Top 50 Most Variable Genes

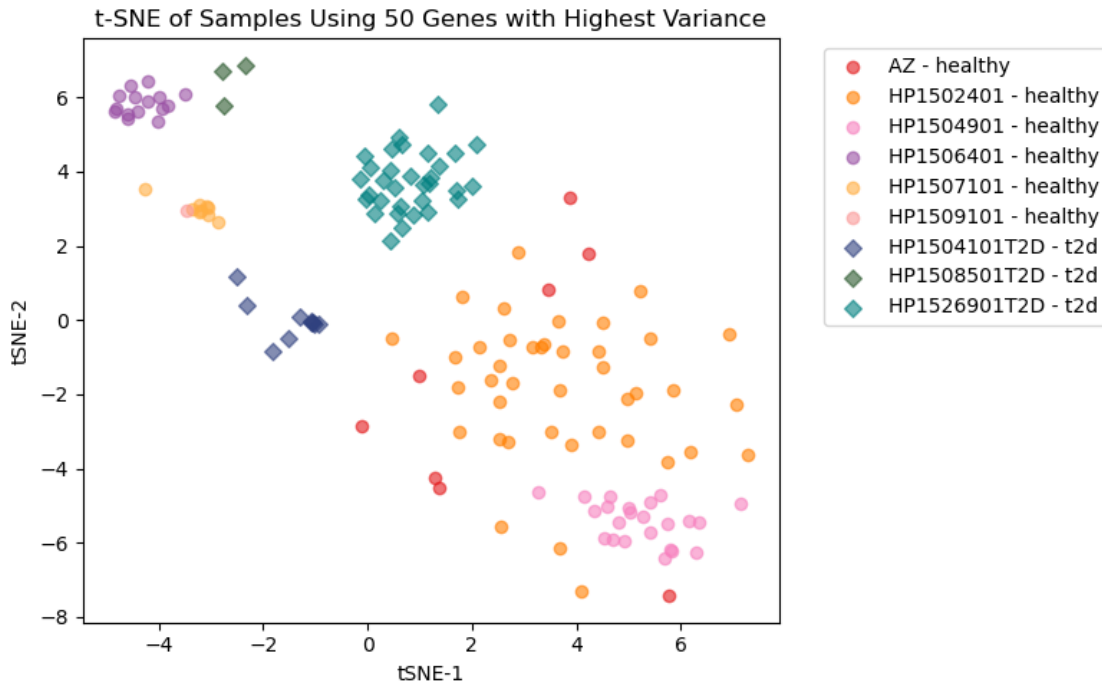
Figure 4B-left

[19]:
```
top_var_genes = X_log.var(axis=0).sort_values(ascending=False).head(50).index
X_top = X[top_var_genes]
X_top.shape
```

[19]: (142, 50)

[20]:
```
X_pca = PCA(n_components=30).fit_transform(X_log)
tsne = TSNE(n_components=2, perplexity=30, random_state=23)
X_tsne = tsne.fit_transform(X_pca)
```

[21]:
```
# Plot with donor-specific colors
plt.figure(figsize=(8, 5))
markers = {'healthy': 'o', 't2d': 'D'}
colors = [
    "#e41a1c",  # warm - red
    "#ff7f00",  # warm - orange
    "#f781bf",  # warm - pink
    "#984ea3",  # warm - purple
    "#ffae42",  # warm - goldenrod
    "#fb9a99",  # warm - salmon

    "#30417f",  # cool - blue
    "#2b592e",  # cool - green
    "#008080"   # cool - teal
]
donors = df_t['donor'].unique()
donor_colors = dict(zip(donors, colors))

for donor in df_t['donor'].unique():
    ix = df_t['donor'].values == donor
    condition = df_t[df_t['donor'] == donor]['condition'].unique()[0]

    plt.scatter(
        X_tsne[ix, 0],
        X_tsne[ix, 1],
        label=f"{donor} - {condition}",
        color=donor_colors[donor],
        marker=markers.get(condition, 'o'),
        alpha=0.6
    )

plt.title('t-SNE of Samples Using 50 Genes with Highest Variance')
plt.xlabel('tSNE-1')
plt.ylabel('tSNE-2')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



1.2.3 Gene Expression Plots

DE genes (Figure 4B-right)

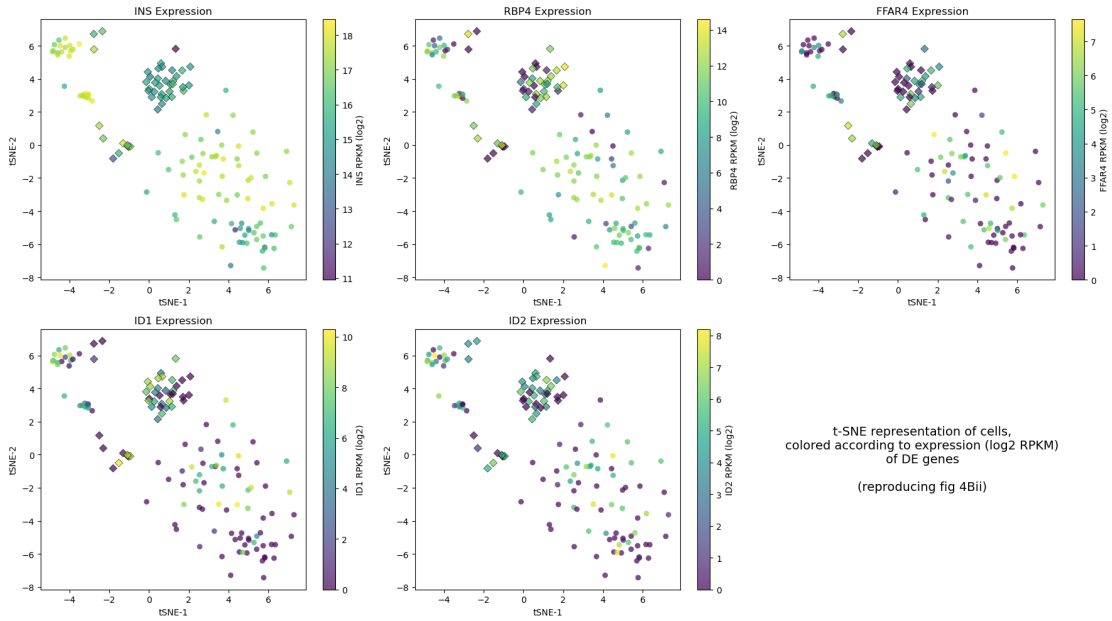
[22]:
```
mg = mygene.MyGeneInfo()

# Query gene symbols
gene_symbols = ['INS', 'RBP4', 'FFAR4', 'ID1', 'ID2']
results = mg.querymany(gene_symbols, scopes='symbol', fields='ensembl.gene',␣
↪species='human')

# Build dict symbol -> Ensembl ID
symbol_to_ensembl_DE = {r['query']: r['ensembl']['gene'] if␣
↪isinstance(r['ensembl'], dict) else r['ensembl'][0]['gene']
                        for r in results if 'ensembl' in r}
```

```
Input sequence provided is already in string format. No operation performed
Input sequence provided is already in string format. No operation performed
```

gene ID3 (ENSG00000283060) is not present in the rpkm_combined.txt, so I omitted it.

[23]:
```
fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()

# Marker logic by condition
condition_markers = {'healthy': 'o', 't2d': 'D'}
```

```
for ax, gene_symbol in zip(axes, symbol_to_ensembl_DE.keys()):
    ensembl_id = symbol_to_ensembl_DE[gene_symbol]
    gene_expr = np.log2(df_t[ensembl_id] + 1)

    # Get min/max for color scale consistency
    vmin = gene_expr.min()
    vmax = gene_expr.max()

    for condition in ['healthy', 't2d']:
        ix = df_t['condition'] == condition
        marker = condition_markers[condition]

        sc = ax.scatter(
            X_tsne[ix.to_numpy(), 0],
            X_tsne[ix.to_numpy(), 1],
            c=gene_expr[ix],
            cmap='viridis',
            s=40,
            marker=marker,
            alpha=0.7,
            vmin=vmin,
            vmax=vmax,
            edgecolor='k' if marker == 'D' else 'none',
            linewidth=0.5
        )

    ax.set_title(f'{gene_symbol} Expression')
    ax.set_xlabel('tSNE-1')
    ax.set_ylabel('tSNE-2')

    # Add colorbar for each subplot
    cbar = fig.colorbar(sc, ax=ax)
    cbar.set_label(f'{gene_symbol} RPKM (log2)')

axes[-1].text(0.4, 0.5,
              't-SNE representation of cells,\ncolored according to expression␣
 ↪(log2 RPKM)\nof DE genes\n\n(reproducing fig 4Bii)',
              ha='center', va='center', fontsize=14)
axes[-1].set_axis_off()

plt.tight_layout()
plt.show()
```



1.3 Feature Selection

[24]:
```
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Extract X (features) and y (labels)
X = df_t.drop(columns=['condition', 'donor'])
X_log = np.log2(X + 1)
y = df_t['condition']

# Encode 'healthy' = 0, 't2d' = 1
le = LabelEncoder()
y_encoded = le.fit_transform(y)
```

[25]:
```
import json

query_cache_file = "query_result.json"

# Batch query: return gene symbol for each Ensembl ID (This will take 2 minutes)
if os.path.exists(query_cache_file):
    with open(query_cache_file, "r") as f:
        query_result = json.load(f)
    print(f"Loaded cached MyGene.info results from {query_cache_file}")
else:
    ensembl_ids = list(X.columns)
```

```
    mg = mygene.MyGeneInfo()
    query_result = mg.querymany(ensembl_ids, scopes='ensembl.gene',␣
 ↪fields='symbol', species='human')

    # Save for future reuse
    with open(query_cache_file, "w") as f:
        json.dump(query_result, f)
    print(f"Fetched MyGene.info results and cached to {query_cache_file}")

# Build a mapping: ENSG ID → gene symbol
ensembl_to_symbol = {
    item['query']: item.get('symbol', item['query'])  # fallback to ENSG ID if␣
 ↪no symbol found
    for item in query_result
}

no_hit_count = sum(1 for item in query_result if item.get('notfound', False))
print(f"{len(ensembl_to_symbol) - no_hit_count} of {len(X.columns)} ensembl to␣
 ↪symbol mapped")
```

```
Loaded cached MyGene.info results from query_result.json
61474 of 62710 ensembl to symbol mapped
```

[82]:
```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded, stratify=y_encoded, test_size=0.3, random_state=23)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train model
#clf = LogisticRegression(penalty='l2', solver='saga', max_iter=1000,␣
 ↪random_state=23)
clf = LogisticRegression(penalty='l1', solver='liblinear', max_iter=1000,␣
 ↪random_state=23) # l1 or l2 penalty
#clf = LogisticRegression(penalty=None, max_iter=1000, random_state=23)

clf.fit(X_train, y_train)
```

[82]:
```
LogisticRegression(max_iter=1000, penalty='l1', random_state=23,
                   solver='liblinear')
```

[83]:
```
feature_importance = pd.Series(clf.coef_[0], index=X_train.columns)
top_genes = feature_importance.abs().sort_values(ascending=False).head(15)
top_genes_named = top_genes.rename(index=lambda ensg: ensembl_to_symbol.
 ↪get(ensg, ensg))
```

```
top_genes_named
```

[83]:
```
TRNM                0.006479
RN7SL1              0.006218
SST                 0.005725
EIF4EBP1            0.003685
GDF15               0.002819
NPTX2               0.002754
CPE                 0.002678
JUNB                0.001988
ENSG00000256148     0.001932
HLA-E               0.001811
SCG2                0.001711
SHISAL2B            0.001105
PCSK1N              0.001099
GPX3                0.000949
H4C3                0.000843
dtype: float64
```

[84]:
```
t2d_linked_genes = ['PHF2', 'GCK', 'SLC30A8', 'IRS1', 'SREBF1',
                     'KCNJ11', 'CDKAL1', 'AZF1', 'GLIS3', 'ZNRF3']

t2d_linked_ensembl = []
for ensembl, symbol in ensembl_to_symbol.items():
    if symbol in t2d_linked_genes:
        t2d_linked_ensembl.append(ensembl)

for ensembl in t2d_linked_ensembl:
    print(feature_importance[ensembl])
```

```
0.0
0.0
0.0
0.0
0.0
0.0
0.0
0.0
0.0
```

[87]:
```
# These are genes commonly-linked to T2D
t2d_genes = [
    "TCF7L2", "CDKAL1", "KCNJ11", "PPARG", "HHEX", "SLC30A8", "IGF2BP2",␣
 ↪"CDKN2A", "CDKN2B",
    "FTO", "MC4R", "CDC123", "PHF2", "SLC16A11", "ANK1", "NOTCH2", "JAZF1",␣
 ↪"WFS1", "GLIS3",
```

```
    "GCK", "GCKR", "HNF1A", "HNF4A", "PAX5", "TET1", "CHL1", "GLRA1", "IAPP",␣
 ↪"CAPN10", "ABCC8",
    "GCGR", "ADAMTS9", "ADIPOQ", "IRS1", "SREBF1", "EXT2", "MEF2C", "TMEM161B",␣
 ↪"CEP120",
    "PRDM6", "STEAP1", "ZNF804B", "ZNRF3", "PRIM1", "IRF2BPL", "LRRC74A"
]

set(top_genes_named.index) & set(t2d_genes)
```

[87]:
```
{'CDKAL1',
 'GCK',
 'GLIS3',
 'IRS1',
 'KCNJ11',
 'PHF2',
 'SLC30A8',
 'SREBF1',
 'ZNRF3'}
```

[ ]:
```
plt.figure(figsize=(10, 4))
colors = ['red' if gene in t2d_linked_genes else 'blue' for gene in␣
 ↪top_genes_named.index]
top_genes_named.plot(kind='barh', color=colors)
plt.gca().invert_yaxis()
plt.xlabel('Feature Importance')
plt.ylabel('Gene Symbol\n(red genes linked to T2D)')
plt.title('Top 15 Predictive Genes (Healthy vs T2D)')
plt.text(0.98, 0.05, "Logistical Regression Model\nL2 Regularization",
         ha='right', va='bottom',
         transform=plt.gca().transAxes,
         bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))

plt.tight_layout()
plt.show()
```

The following genes have been linked to T2D:

['PHF2', 'GCK', 'SLC30A8', 'IRS1', 'SREBF1', 'KCNJ11', 'CDKAL1', 'AZF1', 'GLIS3', 'ZNRF3']

The most predictive gene is IRF2BPL. This gene encodes a transcription factor that may play a role in regulating female reproductive function. [provided by RefSeq, Jun 2012]. It is quite predictive likely because a sex imbalance between health and T2D donors.

M:F donors

5:1 healthy

2:2 T2D

1.3.1 Retrieve Function and Ontology of top predictor genes.

[ ]:
```
gene_names = [gene for gene in top_genes_named.keys()]

mg = mygene.MyGeneInfo()
fields = 'name,symbol,summary,go'

# Query MyGene.info
results = mg.querymany(gene_names, scopes='symbol', fields=fields,␣
 ↪species='human')
len(results)
```

[ ]:
```
def extract_terms(val):
    if isinstance(val, list):
        return '; '.join([v.get('term', '') for v in val])
    elif isinstance(val, dict):
        return val.get('term', '')
    return ''

# Parse results
data = []
for entry in results:
    if 'notfound' in entry and entry['notfound']:
        data.append({'symbol': entry['query'], 'summary': 'Not found', 'GO_BP':␣
 ↪'', 'GO_MF': '', 'GO_CC': ''})
        continue

    go = entry.get('go', {})
    bp = extract_terms(go.get('BP'))
    mf = extract_terms(go.get('MF'))
    cc = extract_terms(go.get('CC'))

    data.append({
        'symbol': entry.get('symbol', ''),
        'name': entry.get('name', ''),
        'summary': entry.get('summary', ''),
        'GO_BP': bp,
        'GO_MF': mf,
        'GO_CC': cc
    })

# Save and print
df = pd.DataFrame(data)
print(df.to_string(index=False))
# df.to_csv('gene_function_ontology.csv', index=False)
# df.to_csv('gene_function_ontology.csv', mode="a", header=False, index=False)
```

1.3.2 Run model with only T2D-linked genes

[ ]:
```
mg = mygene.MyGeneInfo()
results = mg.querymany(t2d_genes, scopes='symbol', fields='ensembl.
 ↪gene,symbol', species='human')
```

[ ]:
```
# Step 2: Extract valid Ensembl gene IDs
ensembl_ids = []
for r in results:
    if 'ensembl' in r:
        if isinstance(r['ensembl'], list):
            ensembl_ids.extend([e['gene'] for e in r['ensembl']])
        else:
            ensembl_ids.append(r['ensembl']['gene'])

ensembl_ids
```

[109]:
```
columns_to_keep = [col for col in ensembl_ids if col in df_t.columns]
t2d_df = df_t[columns_to_keep + ['condition', 'donor']].copy()
```

[96]:
```
from sklearn.ensemble import RandomForestClassifier

# Split data (optional but good practice)
X_train, X_test, y_train, y_test = train_test_split(X, y_encoded, stratify=y,␣
 ↪test_size=0.2, random_state=23)

# Train model
clf = RandomForestClassifier(n_estimators=100, random_state=23)
clf.fit(X_train, y_train)
```

[96]:
```
RandomForestClassifier(random_state=23)
```

[98]:
```
# Extract feature importance
feature_importances = pd.Series(clf.feature_importances_, index=X_train.columns)
top_genes = feature_importances.sort_values(ascending=False).head(20)

# Map feature names to symbols where possible
top_genes_named = top_genes.rename(index=lambda ensg: ensembl_to_symbol.
 ↪get(ensg, ensg))
```

[ ]:
```
plt.figure(figsize=(8, 6))
top_genes_named.plot(kind='barh')
plt.gca().invert_yaxis()
plt.xlabel('Feature Importance')
plt.title('Top 20 Predictive Genes (Healthy vs T2D)')
plt.tight_layout()
plt.show()
```